Stimulus and ActionCable

The Turbo ActionCable helpers are great, but they don't cover every use case for ActionCable we might have. You may need to use ActionCable by using custom JavaScript on the client, either because there is an already existing endpoint or because your task doesn't quite fit with the Turbo patterns. In this section, we're going to use ActionCable to rebuild the "sold out" feature we previously built using polling in Chapter 8, *Talking to the Server*.

To use ActionCable directly, we need to create objects on the client and on the server. On the server, we have *channels* and *connections*. An ActionCable channel is roughly analogous to a Rails controller. The channel is where you put the code that responds to data the server receives over the socket and where you set up the structure of the data being sent to the client. Turbo created a generic channel for us to use; here we're going to build a custom one.

The connection class is a little more abstract. It's used by the channel and typically handles authorization. In setting up our project, Rails created the two top-level classes, `ApplicationCable::Connection` and `ApplicationCable::Channel`, that we can use as base classes. On the client side, we will call a method called `createConsumer`, which returns a data object that we can use to subscribe to a specific server-side channel, and then specify what we want to happen in response to received data.

Let's look at an example of using ActionCable to broadcast sold-out concert data to our schedule page. We'll need to write some server-side code and some client-side code. On the server side, we need to create a channel and also set up the place where data is sent to that channel. On the client side, we need to respond to data when it's sent.

We've set up this functionality to be the most basic ActionCable possible—the data only flows one way from the server to the client, and the identical broadcast goes to all subscribers. The schedule is the same for everybody.

We can start by asking Rails to generate some boilerplate: